

Design and Analysis of Algorithms

CSIT — 5th Semester — 2079 — Answer Sheet
Subject Code: CSC325

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Section A

1. Explain the divide and conquer strategy for problem solving. Describe the worst-case linear time selection algorithm and analyze its complexity.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. Write the dynamic programming algorithm for matrix chain multiplication. Find the optimal parenthesization for the matrix chain product ABCD with size of each is given as A[1][1], B[1][1], C[1][1], D[1][1].

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. What do you mean by Backtracking? Explain the backtracking algorithm for solving 0-1 knapsack problem and find the solution for the problem given below: Items → 1234 Weight (w_i) → 6235 Profit (P_i) → 45107 and capacity of knapsack: 10 kg

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section B

Attempt any Eight questions [8×5=40]

4. Explain the iterative algorithm to find the GCD of given two numbers and analyze its complexity.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Trace the quick sort algorithm for array A[] = {15, 7, 6, 23, 18, 34, 25} and write its best and worst complexity.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Generate the prefix code for "CYBER CRIME" using Huffman algorithm and find the total number of bits required.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Solve the following recurrence relations using masters methoda) $T(n) = 2T(n/2) + kn^2$, $n > 1$, $n = 1$.b) $T(n) = 5T(n/4) + kn$, $n > 1$, $n = 1$.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Find the edit distance between the string "ARTIFICIAL" and "NATURAL" using dynamic programming.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. Find the MST from following graph using Kruskal's algorithm.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Solve the following linear congruences using Chinese Remainder Theorem.
$$\begin{aligned}x &\equiv 1 \pmod{2} \\x &\equiv 3 \pmod{5} \\x &\equiv 6 \pmod{7}\end{aligned}$$

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Define tractable and intractable problem. Illustrate vertex cover problem with an example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Write short notes on:a) Best, Worst and average case complexityb) Greedy Strategy

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.